

PROJECT 02 / DIRECT-TO-CHIP LIQUID COOLING

Designing a 6 kW AI accelerator cooling tray

A model-based workflow linking cold-plate CFD, reduced-order models, manifold hydraulics, and pump/fault decisions.

6.0 kW	9.16 L/min	12.3°C margin
8 × 750 W design load	Modeled pump/tray flow	Below the 85°C project limit

The engineering decision

Retain a 19.05 mm common header as the provisional baseline. At the nominal modeled operating point, the tray reaches 72.72°C estimated peak device temperature with 0.342% flow maldistribution. The result is a design-study prediction, not hardware qualification. [E01–E02]

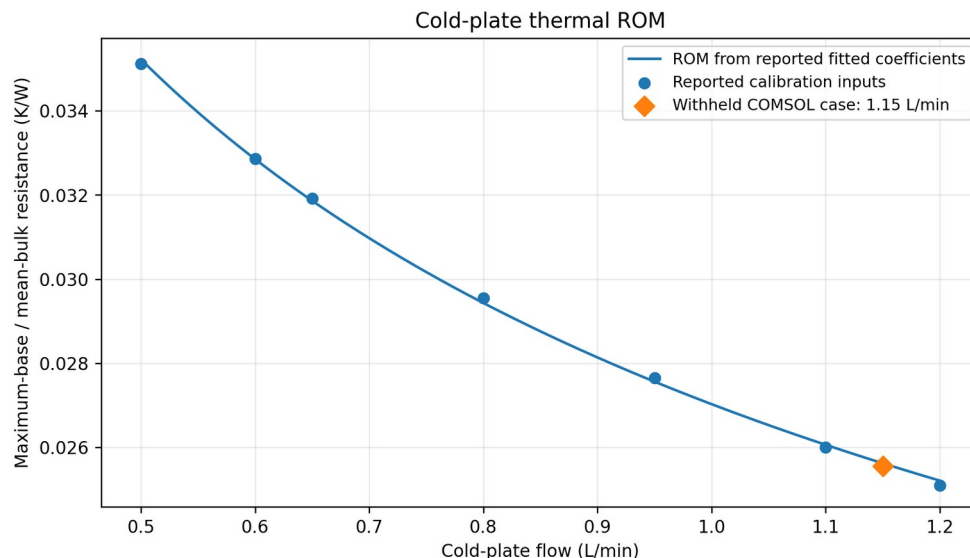
From component physics to a system model

Screened cold-plate candidates in COMSOL; selected a 30-channel, 0.40 × 1.00 mm Candidate C design. Fitted pressure-drop and thermal-resistance ROMs, then integrated branch tubing, illustrative QD/fitting losses, and a scalable same-side manifold network. A representative pump curve determines actual modeled flow rather than prescribing the thermal target. [E10]

Independent component-level CFD check

The 1.15 L/min, 750 W, 30°C COMSOL case was withheld from fitting. The extended calibration flow range is 0.50–1.20 L/min. [E03–E04]

Output	COMSOL	ROM	Difference
Pressure drop	11.258 kPa	11.260 kPa	+0.0178%
Thermal resistance	0.0255542 K/W	0.0256179 K/W	+0.2493%
Maximum base temperature	53.870°C	53.908°C	+0.0382°C



Reported fit and withheld CFD point; resistance is referenced to $(T_{in} + T_{out})/2$. ROM agreement with CFD does not establish hardware accuracy.

Hydraulic limits can fail before temperature

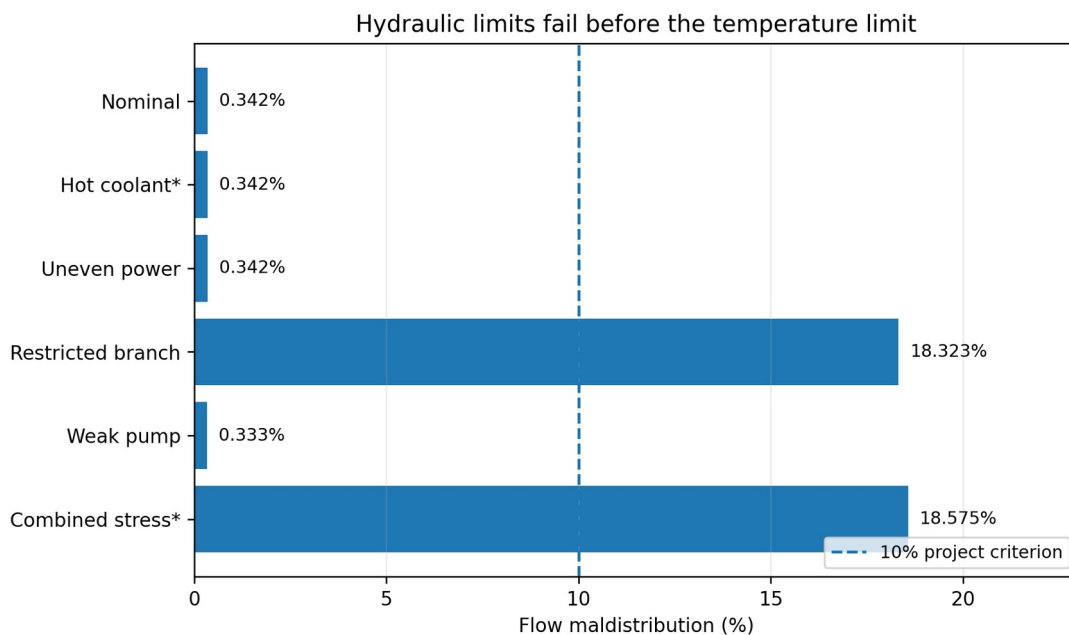
Six reported scenarios separate temperature compliance from flow and uniformity requirements. Nominal tray differential pressure is 11.44 kPa; the illustrative pump provides 6.02% flow margin above the fixed 8.64 L/min target. [E01–E02]

Scenario	Flow L/min	Mflow %	Tmax est. °C	Reported outcome
Nominal	9.1604	0.342	72.72	Pass
Hot coolant*	9.1604	0.342	77.72	Pass*
Uneven power	9.1604	0.342	72.72	Pass
Restricted branch	9.0632	18.323	75.05	Fail: uniformity
Weak pump	8.2056	0.333	74.10	Fail: flow target
Combined stress*	8.1016	18.575	81.67	Fail: flow + uniformity

* 35°C inlet: constant-property sensitivity only, not additional inlet-temperature CFD validation. Mflow = $100 \times (q_{\max} - q_{\min}) / q_{\text{mean}}$. Status uses the adopted fixed-flow and 10% uniformity criteria.

What changed the design judgment?

A 30% increase in one branch hydraulic characteristic raised maldistribution to 18.32% while the estimated device peak remained 75.05°C. The weak-pump case also remained below the temperature limit but missed the flow target. Temperature alone therefore would not identify every operating-requirement violation. [E01]



Fault results replotted from the supplied scenario table. No new CFD or full-system simulation was run to prepare this document.

Scope of the recommendation

The pump curve, QD/fitting coefficients, and 0.025 K/W package-plus-TIM resistance are modeling assumptions. Pressure covers the tray, not a full CDU/facility loop. The 35°C cases are sensitivities; the complete manifold lacks independent CFD/hardware qualification. The 19.05 mm choice is provisional, not a global optimum or released mechanical assembly. [E10]

Evidence: E01 scenario table; E02 pump/tray summary; E03–E04 withheld component comparison; E10 model settings. Full source index and limitations accompany this document in the portfolio package.